



NETAJI SUBHAS INSTITUTE OF TECHNOLOGY, BIHTA

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Session : 2024-2028

Year/Semester: 2nd / 3rd Roll No. 241064

Branch : Computer Science & Engineering

Subject : Digital Electronics

EXTERNAL LAB RECORD

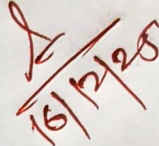
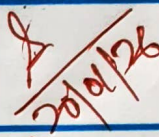
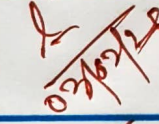
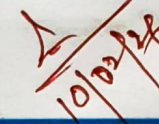
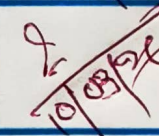


INDEX

Lab Name: Digital Electronics Lab

Faculty Incharge : Mr. Rajni Ranjan

Lab Incharge: Mr. Rakesh Kumar.

Sr. No.	Experiments Name	Date of Experiment	Completion Date	Page No.	Signature	Remarks
1	Realization of All logic gates using Universal gates.	09/12/2025	16/12/2025	1 to 4	 16/12/25	(A+)
2	Realization of Half and Full Adder	06/01/26	20/01/26	5 to 7	 20/01/26	(A+)
3	Realization of Half and Full Subtractor	03/02/26	03/02/26	8 to 10	 03/02/26	(A+)
4	Binary to Gray Code Conversion	10/02/26	10/02/26	11 to 14	 10/02/26	(A+)
5	Realization of Multiplexer & Demux	17/02/26	17/02/26	15 to 16	 17/02/26	(A+)
6	Implementation of Magnitude Comparator	10/03/26	10/03/26	17 to 19	 10/03/26	(A+)
7						
8						
9						
10						

⇒ AIM:- Universal Gates

- (i) Identification and verification of NAND gate (IC #7400) and NOR gate (IC #7402).
- (ii) Construction and verification of all other gate (AND, OR, NOT, X-OR) using.
 - a) Only NAND gate
 - b) Only NOR gate.

⇒ Apparatus:-

Digital Trainer kit, Breadboard, LEDs, IC 7400 (NAND), IC 7402 (NOR), +5V Power Supply, Connecting wires.

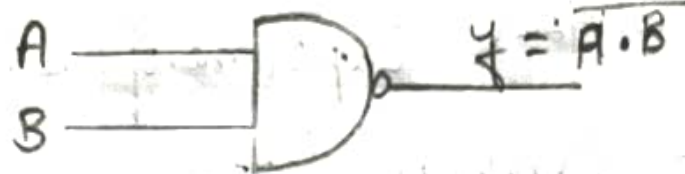
⇒ Theory:-

NAND and NOR gates are called Universal gates because any logic gate can be constructed using only NAND gates or only NOR gates.

(i) NAND Gate:-

The NAND gate represents the complement of the AND operation. Its name abbreviation of NOT AND. The graphic symbol for the NAND gate consists of an AND symbol with a bubble on the output, denoting that a complement operation is performed on the output of the AND gate.

NAND Gate:-
Graphic Symbol:-



NOR Gate Graphic Symbol:-



Truth table of NAND gate.

Input		Output
A	B	$\bar{A} \cdot \bar{B}$
0	0	1
0	1	1
1	0	1
1	1	0

(ii) NOR gate:

The NOR gate represents the complement of the OR operation. Its name is an abbreviation of NOT OR. The graphic symbol for the NOR gate consists of an OR symbol with a bubble on the output, denoting that a complement operation is performed on the output of the OR gate. The truth table and the graphic symbol of NOR gate is shown in the figure.

Truth table of NOR gate.

Input		Output
A	B	$\overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

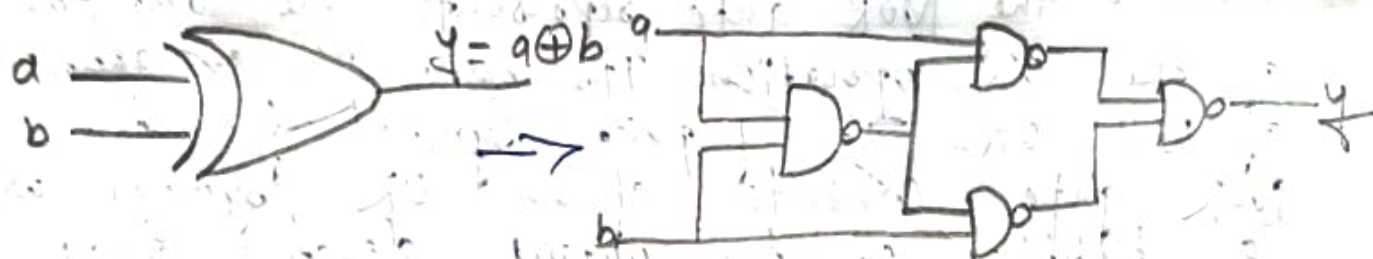
NAND gate as universal gate.

NAND as NOT gate: $a \rightarrow \overline{a}$

NAND as AND gate: $\frac{a}{b} \rightarrow \overline{a \cdot b} \rightarrow a \cdot b$

NAND as OR gate: $a, b \rightarrow y = a + b$

NAND as XOR gate:



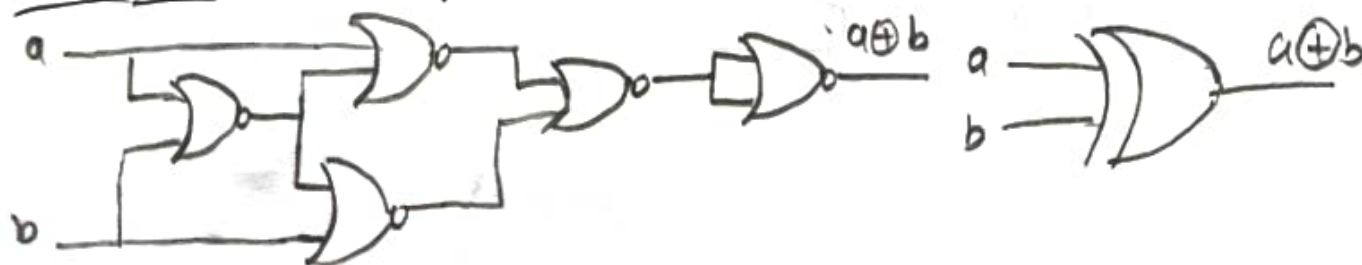
NOR as Universal gate:

NOR as NOT gate: $a \rightarrow \overline{a}$

NOR as OR gate: $\frac{a}{b} \rightarrow a + b$

NOR as AND gate: $\frac{a}{b} \rightarrow a \cdot b$

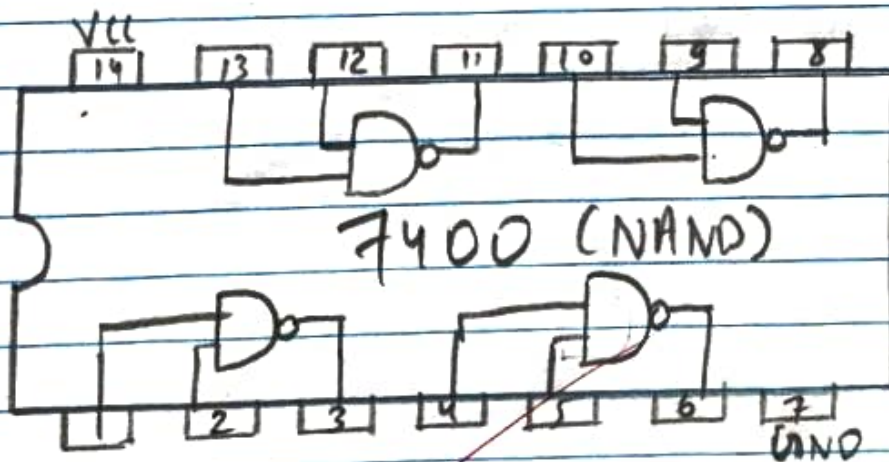
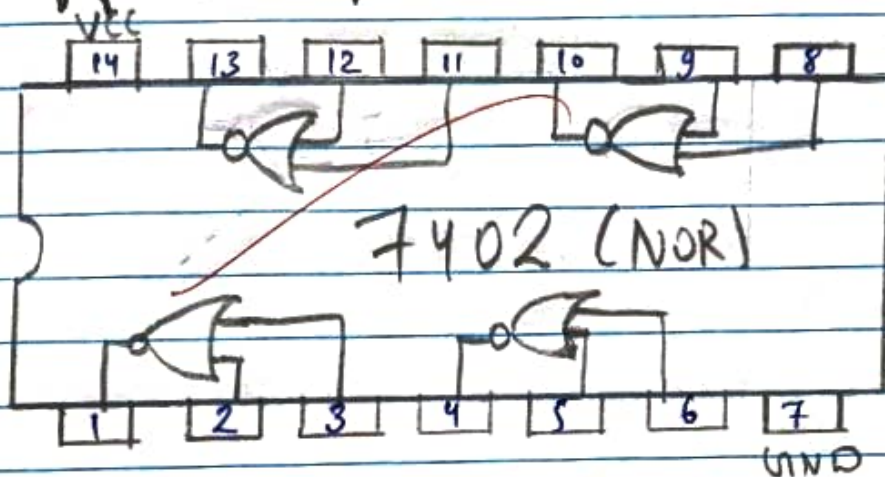
NOR as a XOR gate:



⇒ Truth Table of all the gates:-

Inputs		Outputs					
A	B	AND	NAND	OR	NOR	EXOR	EXNOR
0	0	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	0	1	0	0	1

⇒ Pin Configuration of IC-7402 (NOR), IC-7400 (NAND).



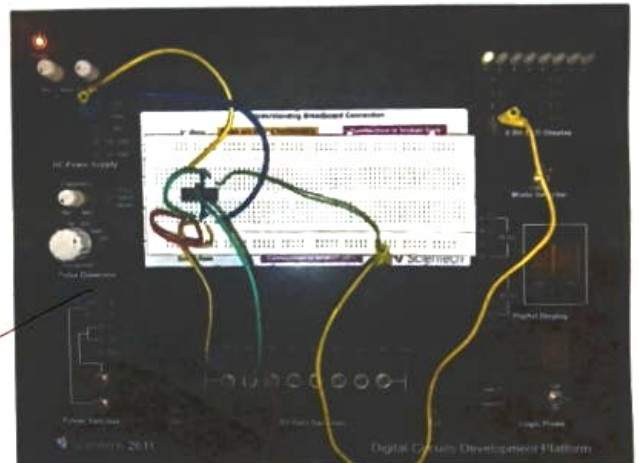
⇒ Procedure:-

- ① Connections are made as per the circuit diagram.

Realize All Gates Using NAND Gate



NAND as AND



NAND as OR



NAND as ExOR

- ② Switch on the Power Supply.
- ③ Apply different combinations of inputs and observe the outputs; compare the outputs with the truth tables.

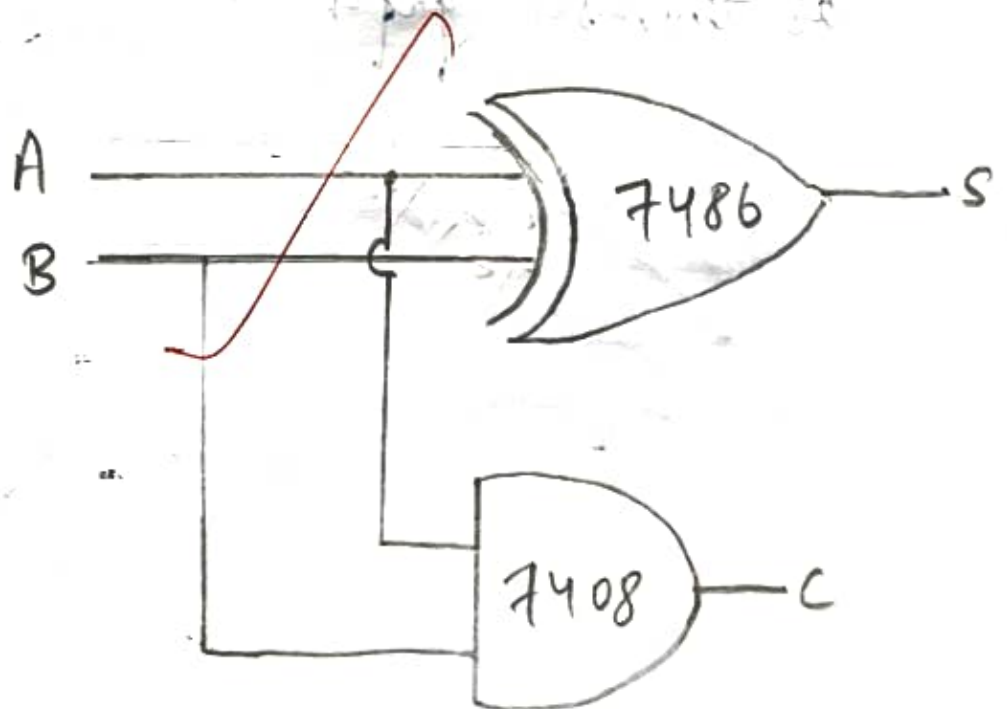
⇒ Precautions:-

- ①. All ICs should be checked before starting the experiment.
- ②. All the connections should be tight.
- ③. Always connect ground first and then the supply.
- ④. Switch off the power supply after completion of the experiment.

⇒ Result:-

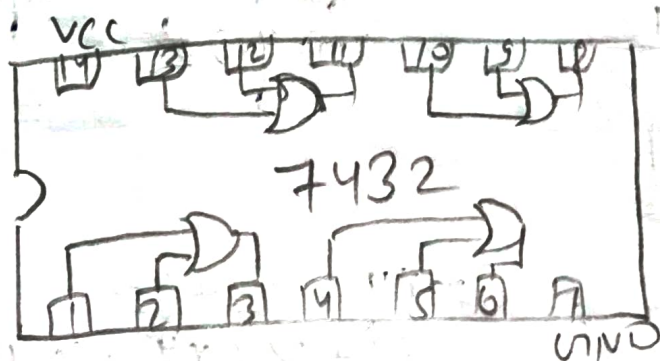
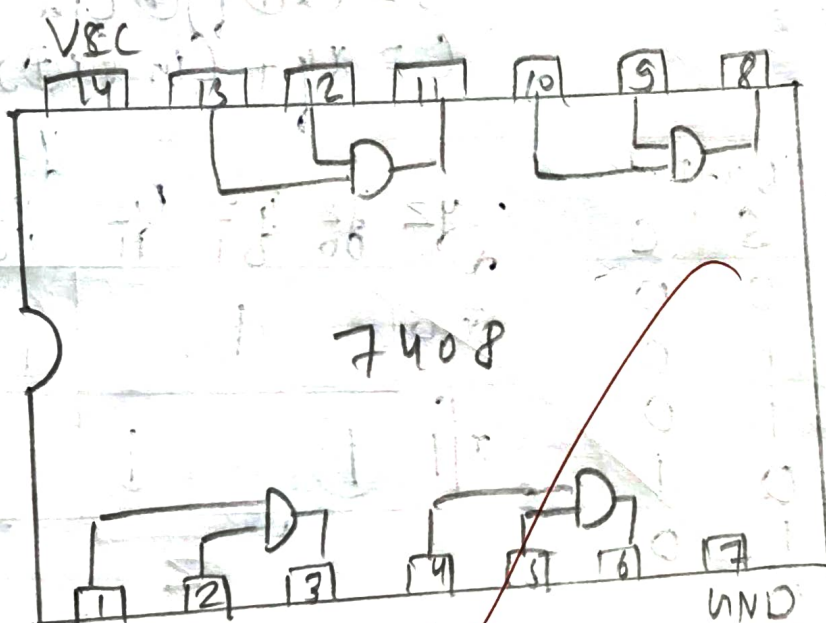
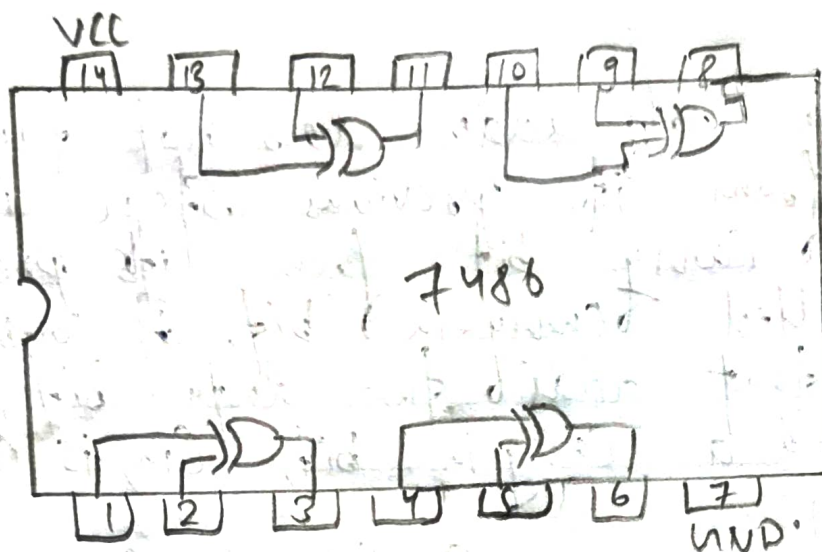
NAND and NOR gates have been studied as universal gates.

Signature
16/12/25



Half Adder.

Pin configuration of IC-7486 (XOR),
IC-7432 (OR), IC-7408 (AND):



Full Adder:-

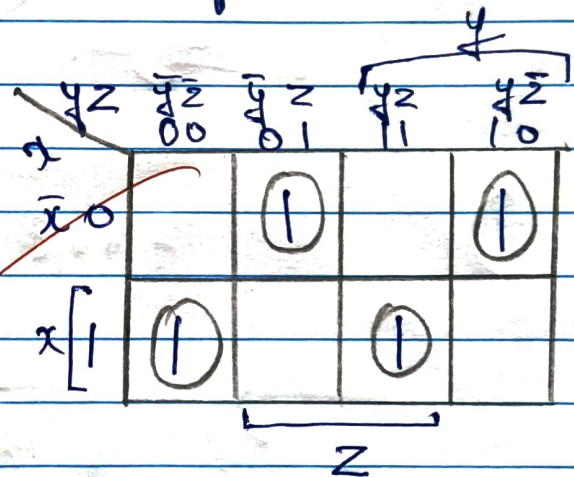
The half-adder does not take the carry bit from its previous stage into account. This carry bit from its previous stage is called carry-in bit. A combinational logic circuit that adds two data bits, A and B, and a carry-in bit, C_{in} , is called a full adder.

$$S = (x \oplus y) \oplus C_{in}$$

$$C = xy + C_{in}(x \oplus y)$$

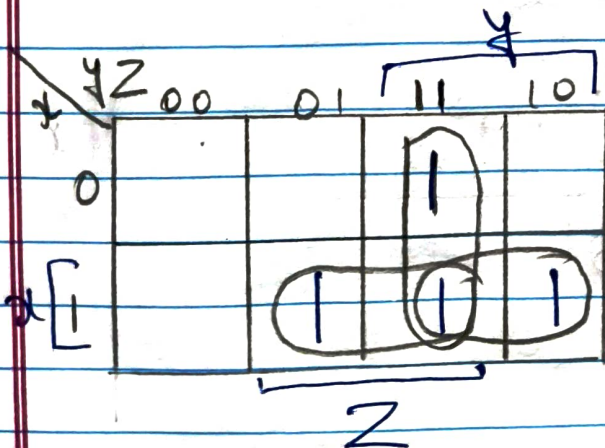
Truth Table.

Inputs			Outputs	
A	B	C_{in}	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$S = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$$

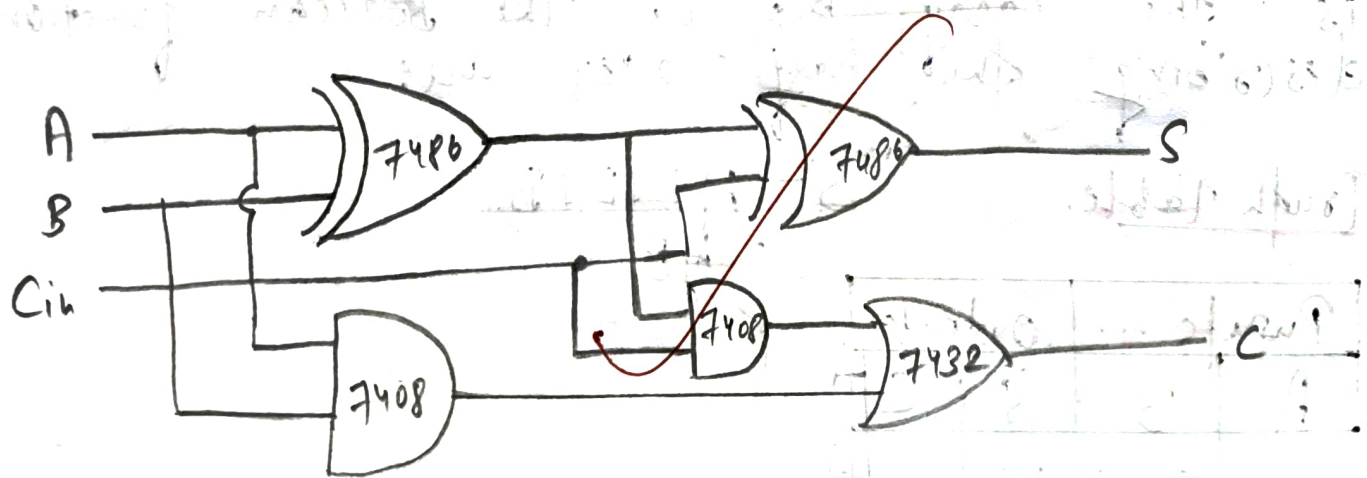
$$= x \oplus y \oplus z$$



$$C = xy + xz + yz$$

$$= xy + z(x\bar{y} + \bar{x}y)$$

$$= xy + z(x \oplus y)$$



Full Adder

Realization of Half Adder and Full Adder

⇒ Aim:-

To realize Half Adder and Full Adder using basic gates.

⇒ Apparatus:-

Breadboard, IC 7486 (XOR), IC 7408 (AND), IC 7432 (OR), LEDs, 5V power supply, connecting wires.

⇒ Theory:-

★ Half Adder:-

A combinational logic circuit that performs the addition of two data bits A and B, is called a half-adder. Addition will result in two output bits; one of which is the Sum bit, S, and the other is the carry bit, C. The Boolean functions describing the half-adder are:

Truth Table.

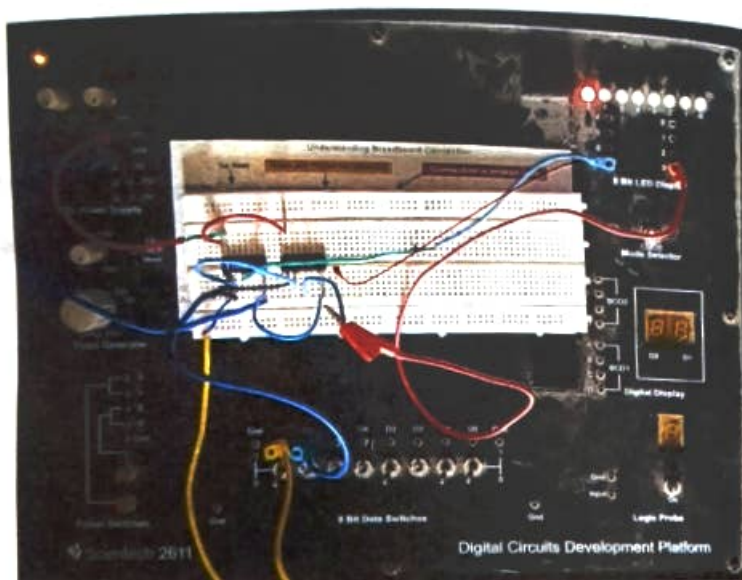
$$S = A \oplus B$$

$$C = AB$$

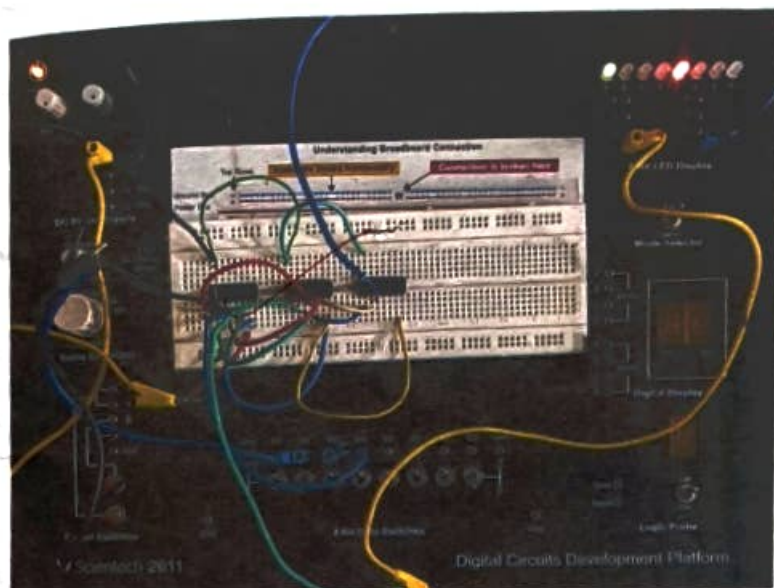
Inputs		Outputs	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Realize Adder Using Logic Gates

Half Adder



Full Adder



✓

Procedure:-

1. Verify the gates.
 2. Make the connections as per the circuit diagram.
 3. Switch on Vcc and apply various combinations of input according to truth table.
 4. Note down the output readings for half adder and full adder.
- ⇒ Sum and the carry bit for different combinations of inputs verify their truth tables.

Precautions:-

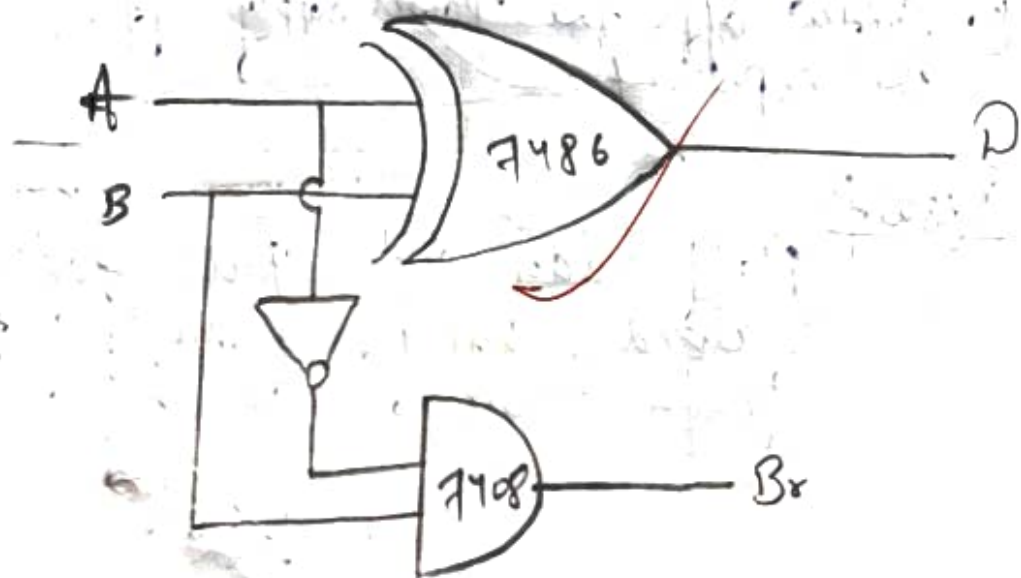
- ① All ICs should be checked before starting the experiment.
- ② All the connection should be tight.
- ③ Always connect ground first and then the supply.
- ④ Switch off the power supply after completion of the experiment.

Result:-

Half adder and Full adder have been realized using basic gates and their truth tables has been verified.

20/01/25

Half
Subtractor



Realization of Half Subtractor and Full Subtractor

⇒ Aim:-

To realize Half Subtractor and full Subtractor.

⇒ Apparatus:-

Breadboard, IC 7486 (XOR), IC 7408 (AND), IC 7404 (NOT), IC 7432 (OR), LEDs, 5V Power Supply, connecting wires.

⇒ Theory:-

Half Subtractor:-

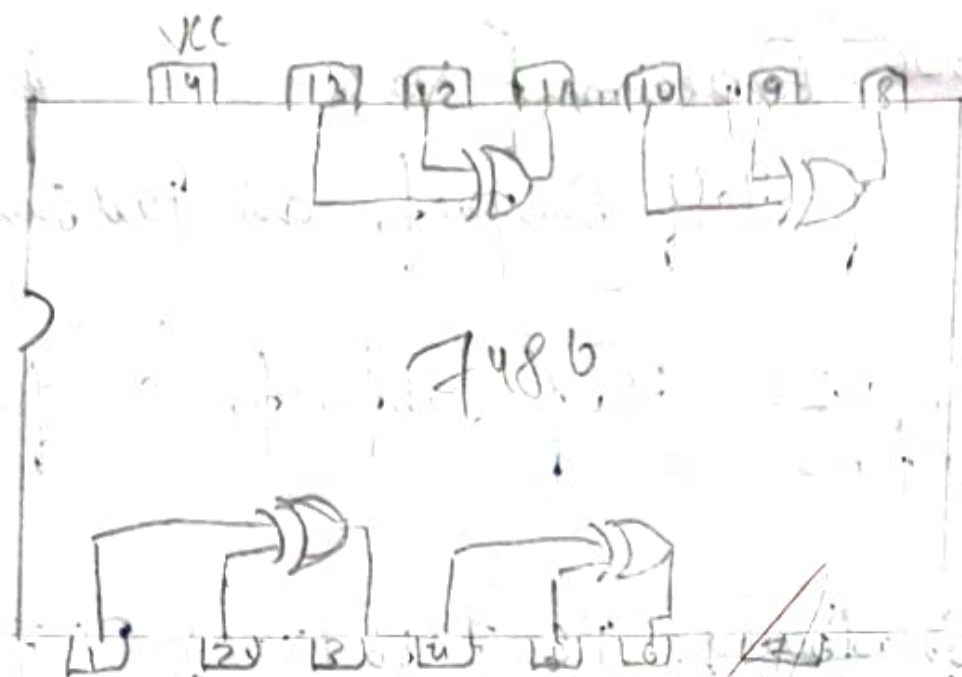
Subtracting a single-bit binary value B from another A (i.e. $A-B$) produces a difference bit D and a borrow out bit B-out. This operation is called half subtraction and the circuit to realize it is called a half Subtractor. The Boolean functions describing the half-Subtractor are:

$$S = A \oplus B$$

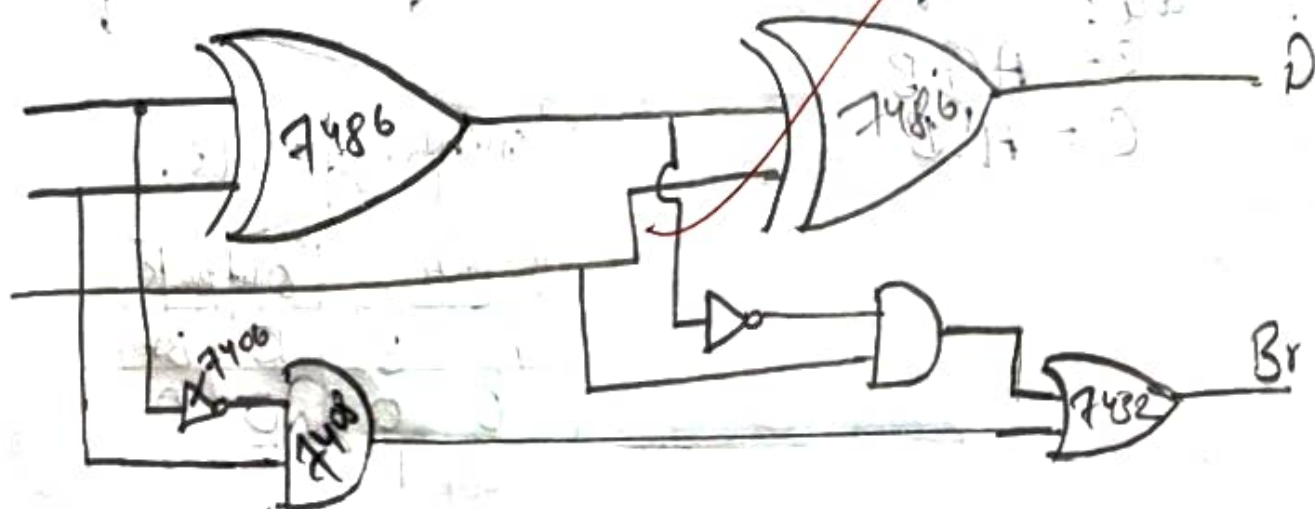
$$C = A' \cdot B$$

Truth table.

Inputs		Outputs	
A	B	D	Br
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



Pin Diagram: ~~7486~~ XOR



Full Subtractor

Full Subtractor:-

Subtracting two single-bit binary values, B , C_{in} from a single-bit value A produces a difference bit D and a borrow out B_r bit. This is called Full Subtraction.

The Boolean functions describing the full-subtractor are:-

$$D = (x \oplus y) \oplus C_{in}$$

$$B_r = A'B + A'(C_{in}) + B(C_{in})$$

Truth Table.

Inputs			Outputs	
A	B	C_{in}	D	B_r
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$x \backslash yz$	00	01	11	10
0		1		1
1	1		1	

z

$$S = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy z$$

$$= x \oplus y \oplus z$$

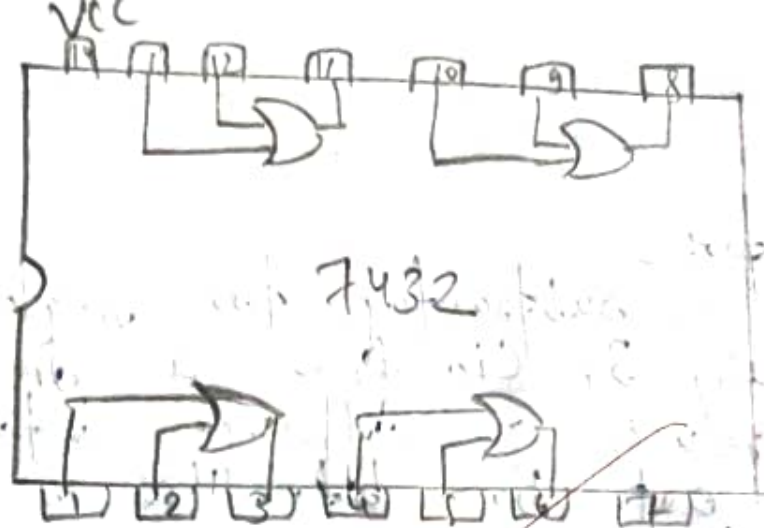
$x \backslash yz$	00	01	11	10
0			1	
1		1	1	1

z

$$C = xy + xz + yz$$

$$= xy + z(x\bar{y} + \bar{x}y)$$

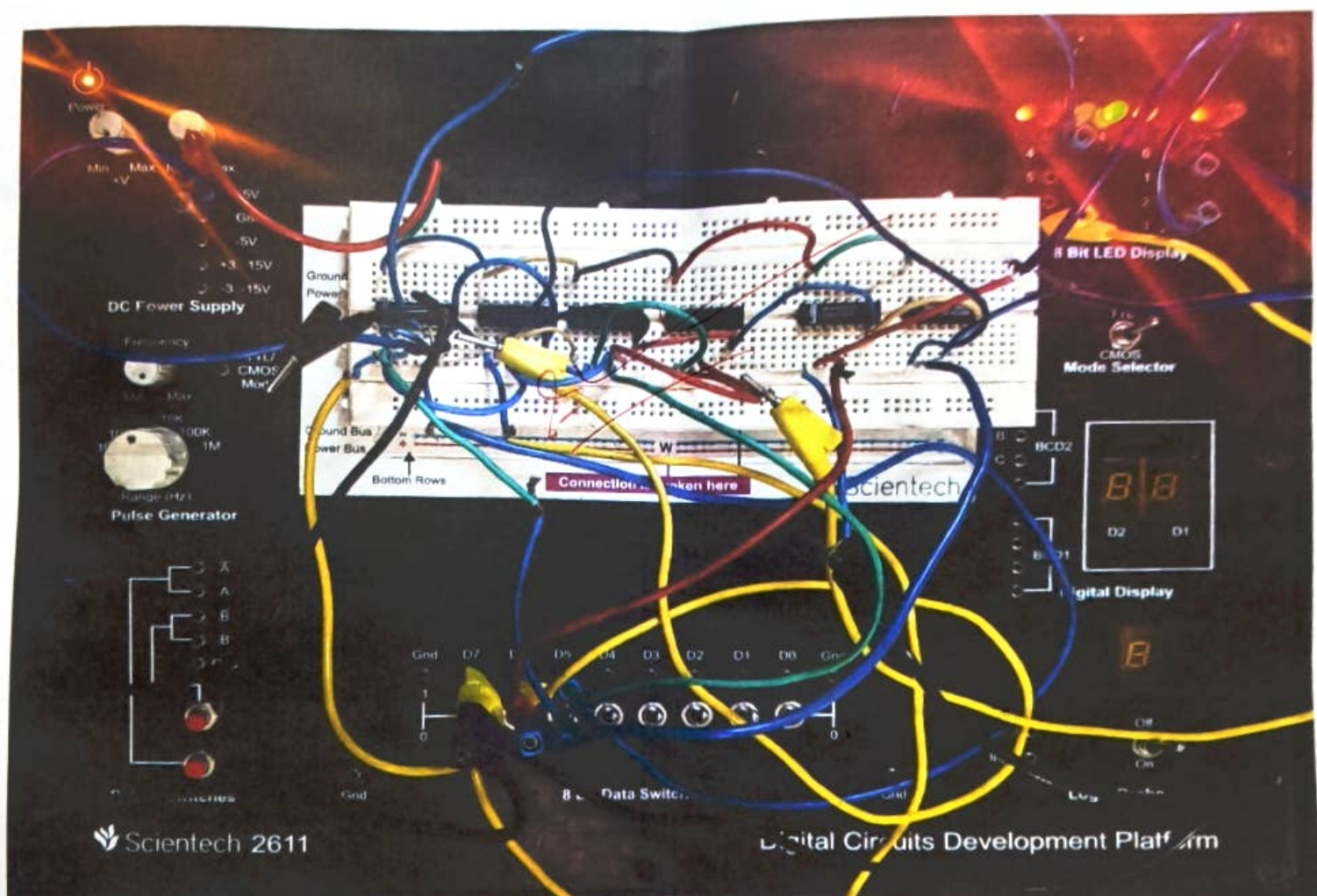
$$= xy + z(x \oplus y)$$



Pin Diagram of 7432 (OR gate)

Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1

Realize Half and Full Subtractor Using Logic Gate



Procedure:-

- * Check all the components for their working.
- * Insert the appropriate IC into the IC base.
- * Make connections as shown in the logic diagram.
- * Verify the results and observe the outputs.

Precautions:-

- * All ICs should be checked before starting the experiment.
- * All the connections should be tight.
- * Always connect ground first and then the supply.
- * Switch off the power supply after completion of the experiment.

Result:-

Half Subtractor and full Subtractor have been studied and their truth table has been verified.

[Signature]
07/02/23

Binary to Gray Code Conversion.

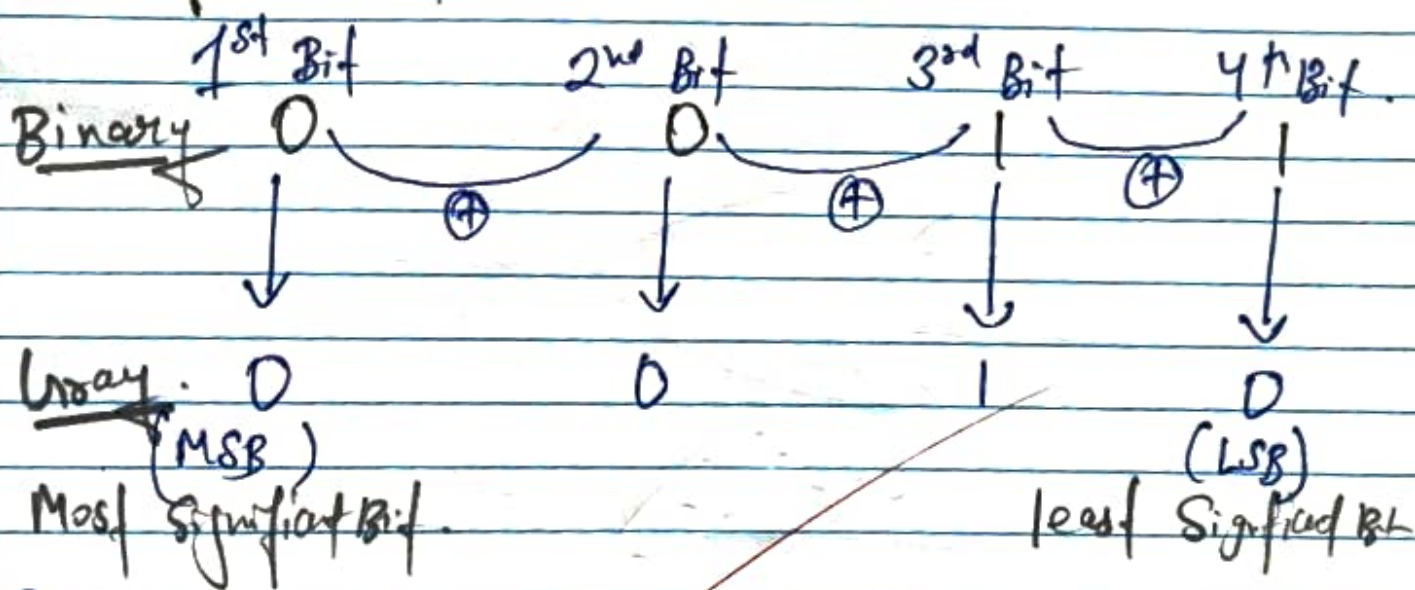
⇒ AIM:-

To verify & realize Binary to Gray code Conversion using XOR gate Binary Kit.

Toolkit used:-

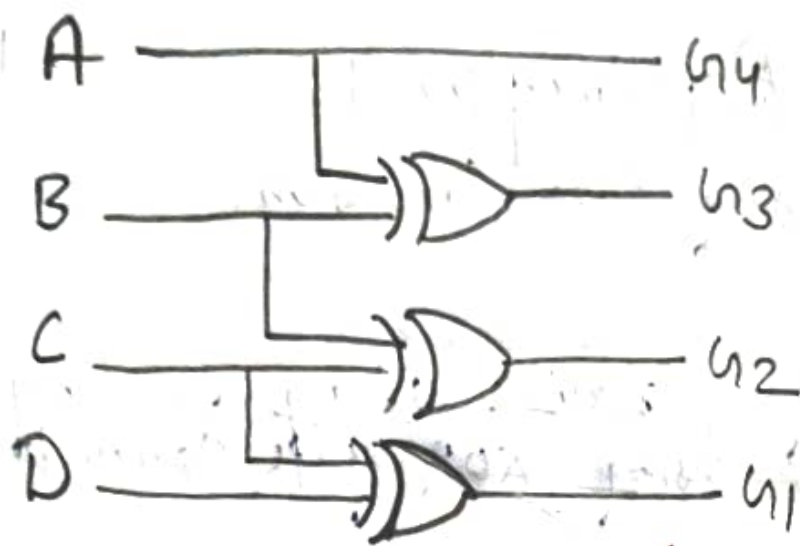
1. DBO6 Binary to Gray code Converter.
2. Digital trainer kit with input power supply.
3. +5V DC Power Supply adapter.
4. Patch Cords.

Theory:-

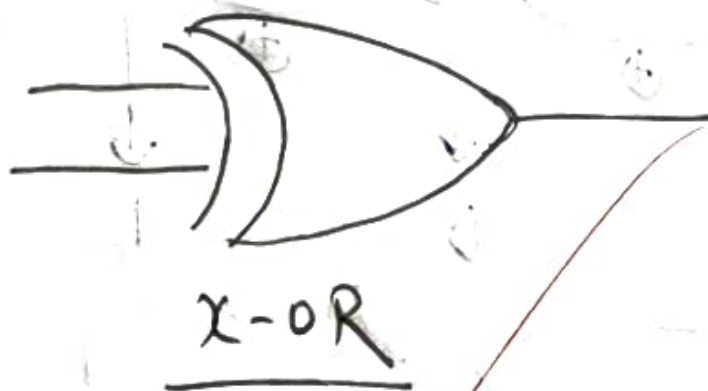


⇒ For Binary to Gray Code Conversion using XOR gates from figure 1.

- ① The MSB of the Gray code will be 0, same as of the MSB of Binary code is 0.
- ② Now, the second gray bit will be XOR of 1st & 2nd binary bits as, $0 \oplus 0 = 0$.
- ③ Similarly, third gray bit will be XOR of 2nd



Binary to Gray code Conversion
Circuit Diagram



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

6th 3rd binary bits as, $0 \oplus 1 = 1$

4. And Lastly, $\rightarrow$ LSB gray bit will be XOR of 3rd and 4th binary bits as $1 \oplus 1 = 0$

5. Therefore the equivalent Gray Code for the given binary code 0011 will be 0010.

Truth table of Binary to Gray Code Conversion

Binary Code				Gray Code			
B3	B2	B1	B0	G3	G2	G1	G0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Code Converter: Binary-Gray and Gray-Binary



Connections:-

- 1.) Connect +5V pin from the trainer on the left side to +5V pin on the DB06 using a patch cord wire.
- 2.) Connect GND pin from the trainer to ground symbol pin on the DB06.
- 3.) Connect the input section pins no. D3, D2, D1 and D0 on the bottom of the trainer to B0, B1, B2 and B3 pins of the DB06 binary to gray code circuit on the left of the panel respectively.
- 4.) Connect the pin no. G0, G1, G2 and G3 of DB06 binary to gray code circuit to the output pin no. D3, D2, D1 and D0 of the 8-bit LED display of the trainer respectively, in order to display the gray code from the binary inputs.
- 5.) Make sure all your connections are right then turn on the power supply.
- 6.) Turn D3 Switch to position 1 and keep D0, D1 and D2 switches at position 0 then observe the output on 8-bit LED display (green light indicates 0 and red light indicates 1).

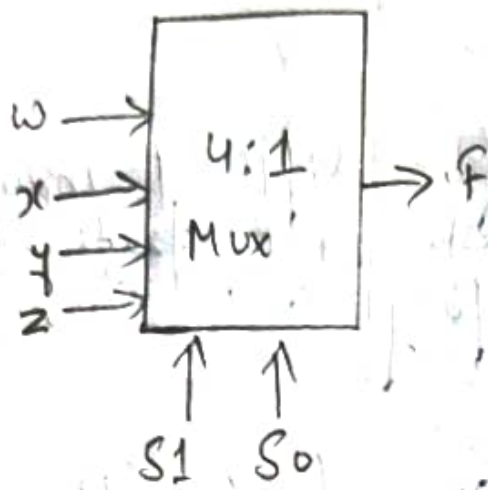
7) Observe the output for different input combination as shown in truth table of binary to gray code conversion on table (1).

8) verify truth table on table (1).

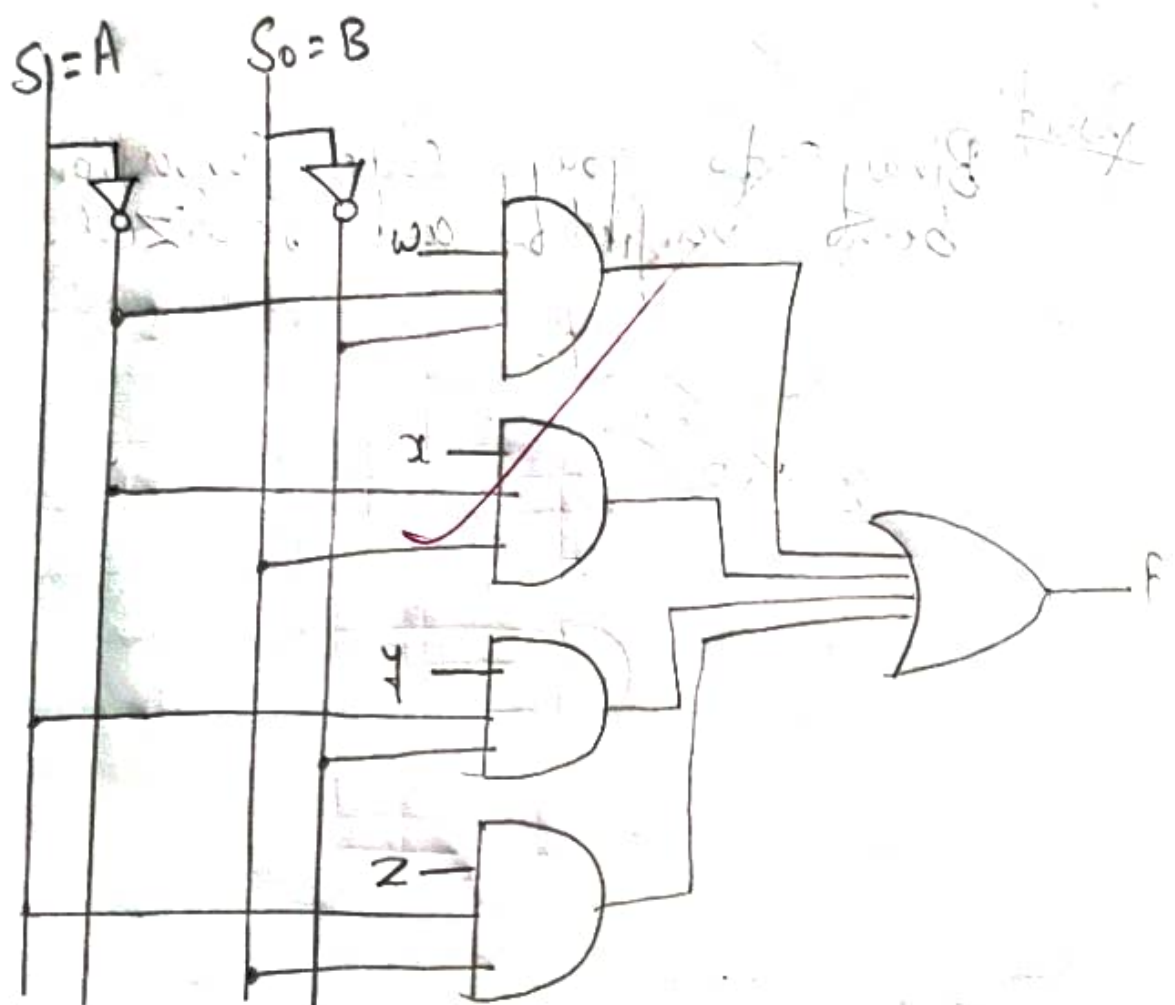
Result:

Binary to gray code conversion has been verified and realized.

10/07/24



Select	Input	Output
S_1	S_0	
0	0	w
0	1	x
1	0	y
1	1	z



Logic Diagram and Truth Table of 4:1 Mux

Realization of Multiplexer and Demultiplexer.

⇒ AIM:-

To realize multiplexer and demultiplexer using logic gates.

⇒ Apparatus:-

Breadboard, IC 7400 (NAND), IC 7410 (3 input NAND), LEDs, 5V power supply, connecting wires.

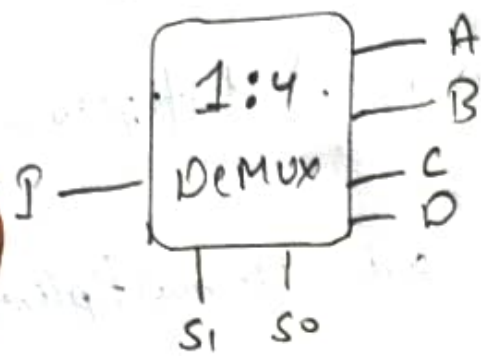
⇒ Theory:-

Multiplexer (Mux):-

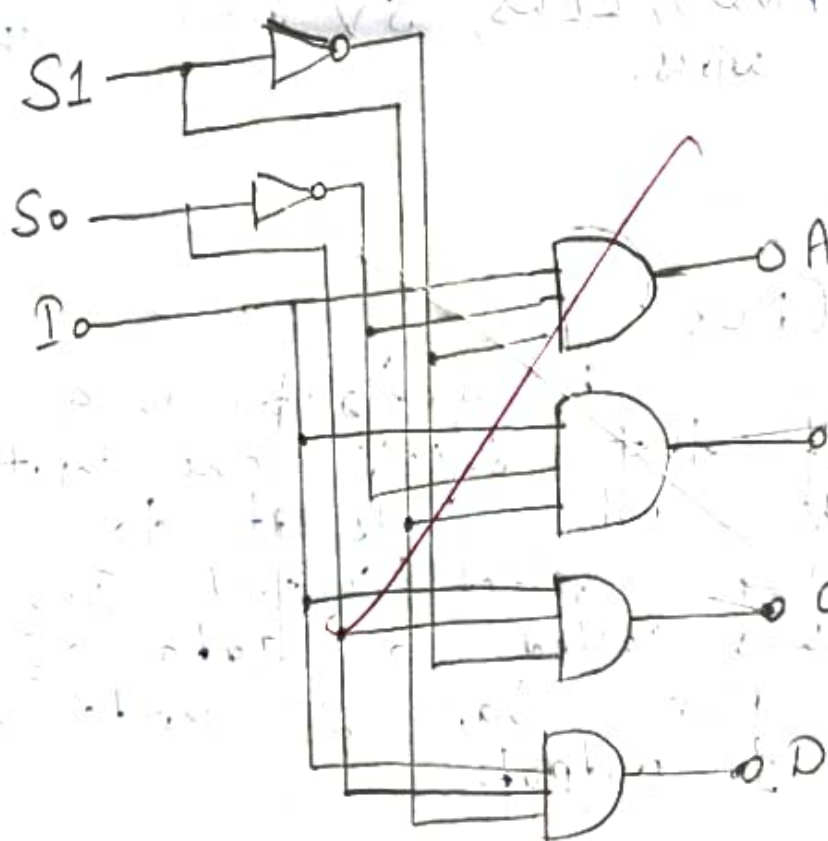
A multiplexer is a many-to-one circuit that selects one input from many inputs and sends it to a single output using select/control signals. It is also called a data selector. A general Mux has 2^n inputs, n select lines, and 1 output.

De-Multiplexer (DEMux):-

A de-multiplexer is a one-to-many circuit that takes one input and sends it to one of many outputs based on select/control signals. A general Demux has 1 input, n select lines, and 2^n outputs, and it can be implemented using a decoder with enable.

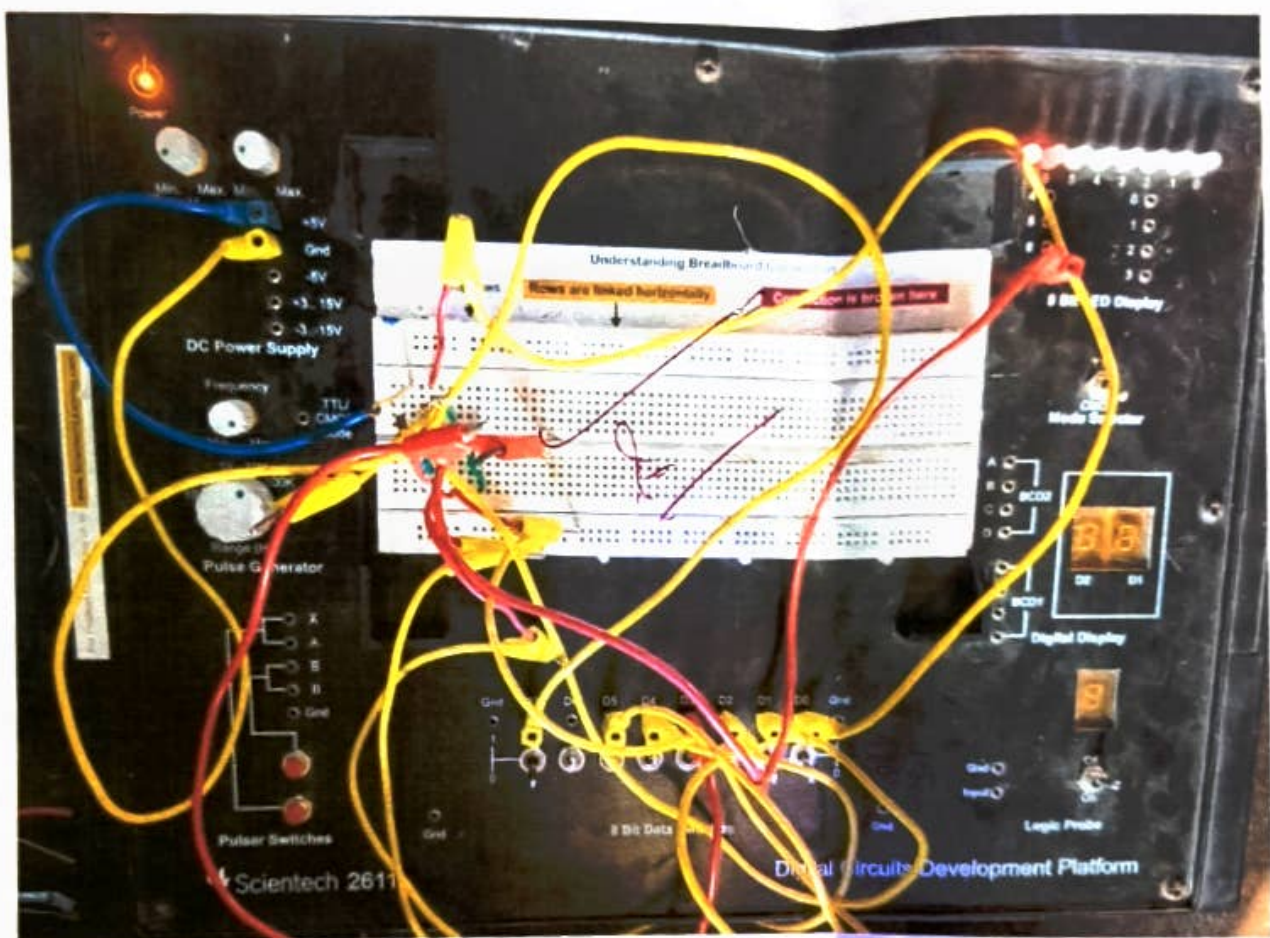


I	Select		Output			
	S ₁	S ₀	A	B	C	D
1	0	0	1	0	0	0
1	1	0	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	1



Logic Diagram and Truth Table of 1:4 Demux

DE-MULTIPLEXER



Procedure:-

- * Check all the components for their working.
- * Insert the appropriate IC into the IC base.
- * Make connections as shown in the circuit diagram.
- * Verify the results and observe the outputs.

Precautions:-

All ICs should be checked before starting the experiment.

1. All the connections should be tight.
2. Always connect ground first and then the supply.
3. Switch off the power supply after completion of the experiment.

Result:-

Multiplexers and Demultiplexers have been studied and their truth table has been verified.

[Signature]
17/02/26

AIM:-

To design & implement a Single bit/ two-bit digital Comparador Circuit.

Apparatus Required:-

1. AND Gate IC 7408
2. X-OR Gate IC 7486
3. OR Gate IC 7432
4. NOT Gate IC 7404
5. 4-Bit Magnitude Comparator IC 7485
6. Bramer Kit.
7. Patch Cords.

Theory:-

A magnitude Comparator is a Combinational Circuit used to compare two binary numbers A and B and determine their relative size. It gives three possible outputs: $A > B$, $A = B$, or $A < B$.

$$A = A_3 \ A_2 \ A_1 \ A_0$$

$$B = B_3 \ B_2 \ B_1 \ B_0$$

Comparator is specified by three binary variables that indicate whether $A > B$, $A = B$ (or) $A < B$.

$$A = A_3 \ A_2 \ A_1 \ A_0$$

$$B = B_3 \ B_2 \ B_1 \ B_0$$

The equality of the two numbers A and B is displayed in a combinational circuit designated by the symbol $(A=B)$. This indicates that if A is greater than B , then inspect the relative magnitude of pairs of significant digits starting from most significant position. If A is 0 and that of B is 0, we have $A=B$. The sequential comparison can be expanded as:

$$A > B = A_3 B_3^1 + x_3 A_2 B_2^1 + x_3 x_2 A_1 B_1^1 + x_3 x_2 x_1 A_0 B_0^1$$

$$A < B = A_3^1 B_3 + x_3 A_2^1 B_2 + x_3 x_2 A_1^1 B_1 + x_3 x_2 x_1 A_0^1 B_0$$

The same circuit can be used to compare the relative magnitude of two BCD digits.

A/A ₀	B/B ₀			
	00	01	11	10
00				
01	1			
11	1	1		1
10	1	1		

$$A > B = A_0 \bar{B}_0 B_1 + A_1 \bar{B}_1 + A_1 A_0 \bar{B}_0$$

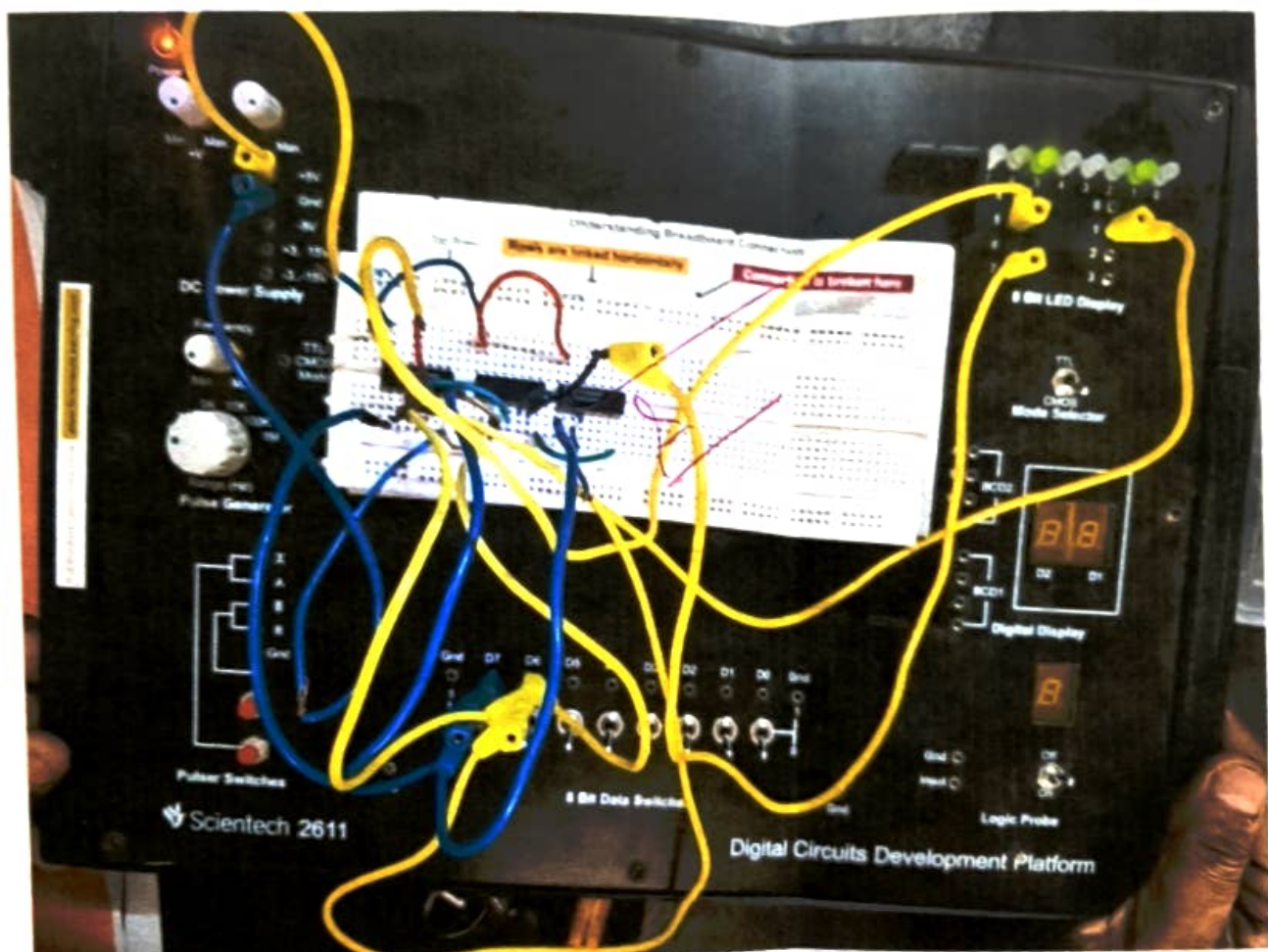
A/A ₀	B/B ₀			
	00	01	11	10
00			1	1
01			1	1
11				
10			1	

$$A < B = \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_0 B_1 + \bar{A}_1 B_1$$

A/A ₀	B/B ₀			
	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$A = B = (B_0 \bar{B}_0)(A_1 \bar{B}_1)$$

COMPERATOR



A ₁	A ₀	B ₁	B ₀	A > B	A = B	A < B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

Procedure:-

- * Connections are given as per Circuit diagram.
- * Logical inputs are given as per Circuit diagram.

Result:-

Observe the output and verify the truth table.

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